

Week 10 – String Algorithms

■ Learning Objectives

- Master string manipulation methods and algorithms.
- Apply searching, splitting, joining, and replacing operations.
- Build algorithms for palindrome and anagram problems.
- Practice parsing and transforming text efficiently.

■ Key Concepts

1. Strings are immutable – operations return new copies.
2. Common functions: `len()`, `find()`, `replace()`.
3. Splitting & Joining: `split()` and `' '.join(list)`.
4. Checking substrings: `if sub in text`.
5. Iterating through strings with loops for pattern matching.

Example 1 – Reverse a String

```
text = "hello"  
print(text[::-1]) # 'olleh'
```

Example 2 – Count Vowels

```
s = "programming"  
vowels = "aeiou"  
count = 0  
for ch in s:  
    if ch in vowels:  
        count += 1  
print(count)
```

Example 3 – Palindrome Check

```
word = input("Enter word: ").lower().replace(" ", "")
if word == word[::-1]:
    print("Palindrome")
else:
    print("Not palindrome")
```

Example 4 – Anagram Check

```
a = input("Word 1: ").lower().replace(" ", "")
b = input("Word 2: ").lower().replace(" ", "")
if sorted(a) == sorted(b):
    print("Anagram")
else:
    print("Not Anagram")
```

■ Practice Problems

Practice 1 – Replace Vowels with *

```
text = input("Enter text: ")
vowels = "aeiouAEIOU"
for v in vowels:
    text = text.replace(v, "*")
print(text)
```

Practice 2 – Word Frequency Counter

```
text = input("Enter sentence: ").lower()
words = text.split()
freq = {}
for w in words:
    freq[w] = freq.get(w, 0) + 1
for w in sorted(freq):
    print(w, ":", freq[w])
```

Practice 3 – Extract Numbers from String

```
s = "abc123xyz45"
num = ""
for ch in s:
    if ch.isdigit():
        num += ch
print("Numbers found:", num)
```

■ Homework Challenge

Write a program that reads text input and outputs each unique word and its frequency in descending order of frequency.

Expected Solution:

```
text = input("Enter text: ").lower()
words = text.split()
freq = {}
for w in words:
    freq[w] = freq.get(w, 0) + 1
for w, f in sorted(freq.items(), key=lambda x: -x[1]):
    print(w, ":", f)
```